Finding the market value of your source code

(English translation of Author's manuscript in Japanese by COPILOT)

Author: **Hideo Watanabe**—just a Japanese individual



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1)**Introduction : Genius Furomuda**

People who come from manufacturing often feel a strange sense of inferiority toward those in the IT industry—as if everyone there were a genius. I was no exception.

About twenty years ago, I read a blog that began with the striking line: “Every single line of source code is a management decision.”
It was from the wildly popular blog series, *Bunretsu Kanchigai-kun Theater* by Furomuda.
At that time, I didn’t even know what source code was. While reading, ‘Source code must be the equivalent of innovation seeds—the technology seeds—in manufacturing, I thought:
Why did I think that?
Let me borrow a passage from Furomuda’s post dated January 18, 2006, titled “**Programming Is the Accumulation of Management Decisions**.”
It is still available today.
He wrote:

Software development is fundamentally a core managerial act that resists outsourcing, yet at the same time requires such a high level of specialization that outsourcing becomes unavoidable. This contradiction defines the profession of software engineers.
The only way to resolve this contradiction is to have people who possess both software expertise and business expertise design software architecture and business architecture as one inseparable whole. These people are often called “hybrid talent,” though the term is not quite

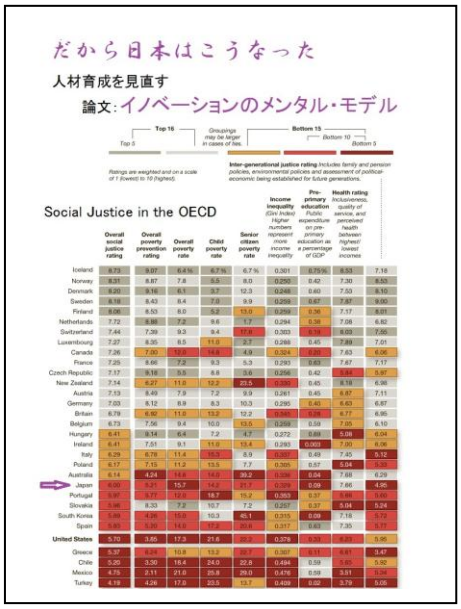
accurate. They do not simply combine two skills; they operate in a domain where technology and business are fused.
When you talk with them, they effortlessly connect bit-level circuit operations to the performance of a device, to the new products it enables, to the users it creates, to the size of the market—hundreds of billions of yen—and even to the kinds of talent and companies needed for development. They are not switching between business and technology; they are performing a single act that inherently contains both.
(end quote)

From the part beginning with “differences in bit-level circuit operations,” I realized that his thinking was exactly the same as developing new products from technology seeds in manufacturing.
And when he casually extended the discussion to “creating a market worth hundreds of billions of yen,” I was stunned.
Was he actually inferring the upper limit of the market value of source code?

Around 2000, I received a request from someone planning a venture capital fund:
“Can you create an evaluation guideline for management teams?”
I produced quite a detailed guideline.
The reason I was stunned was simple: I had already concluded that the most critical factor is whether the management team can infer the upper limit of the market value of their technology seeds.
The guideline was originally intended for VC due diligence.
Later, I reversed its perspective and rewrote it as a framework for

developing innovation talent. This eventually became my 2012 electronic publication on Puboo, *The Mental Model of Innovation: Why Japan Ended Up This Way*, authored under my nickname HideoJapan.

https://puboo.jp/book/38523



By 2006, the idea of inferring upper limits was also being discussed at MIT OpenCourseWare (OCW).

I introduced some of that content in the paper, though their work had not yet reached a concrete methodology.

According to Wikipedia, source code is plain text written by humans—not generated or transformed by machines—and represents the actual substance of software. As the word “source” suggests, software runs on machine code compiled from source code. In that sense, source code is the origin of software. If software is considered a product, then source code corresponds to the technology seeds in manufacturing.

But since I am not from the IT world, I cannot assert this with confidence. That is why I decided to publish this booklet in English.

I actually asked Copilot to translate my manuscript in Japanese under one condition:

“I don’t need polished English. As long as the warmth of the author comes through, that’s enough.” Because this book is a question—**a call for dialogue**.

Whether in Silicon Valley, India, or Europe, anywhere is fine.

If there is a method to infer the upper limit of the market value of source code, I want to know.
In return, I will disclose here the method used in manufacturing to infer the upper limit of technology seeds.

A clumsy belief but not a joke: Sharing wisdom makes the world peaceful. Right?

For anyone leading innovation, the ability to infer the upper limit of technology seeds is an essential mental model—especially in the early stage of development, before even a prototype exists.
How much could this possibly sell?
Even a hypothesis is fine. Without it, the project cannot move forward.
You cannot know how much investment is justified.
This applies both to raising funds from venture capital and to internal corporate projects.

Isn't this the same for traditional software source code and for source code in the Physical AI era?
I am not sure. I only know manufacturing.
Conversely, many people in IT seem unfamiliar with the realities of manufacturing.
I have reasons for thinking so—because I have been genuinely shocked by certain statements. If someone demands evidence, I can show it.

And when it comes to AI robotics, isn't that manufacturing itself?

You show AI endless footage of master craftsmen, letting it absorb patterns.

You link tacit knowledge to software development through AI's predictive power.

I understand the intention. But on both sides—IT and manufacturing—not everyone is the kind of person Furomuda described:

someone who performs a single act that is simultaneously business and technology.

Furomuda says such people exist outside the three-sigma range—about 0.1%, one in a thousand. He calls them "three-sigma people."

They are rare. I will return to this later, but the preface is getting long, and I'm starting to annoy myself.

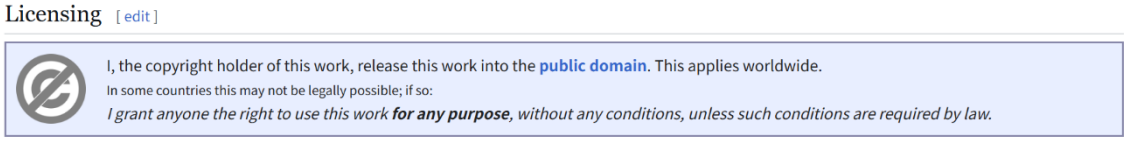
So let me move on and introduce the method for inferring the market value upper limit of technology seeds in manufacturing.

2) Upper Limit of Market Value for Technology Seeds

Many readers have probably seen the diagram that combines a normal distribution with an S-curve—the innovation diffusion process proposed by sociologist Everett Rogers in 1962.

It shows that the upper limit of the S-curve corresponds to the area under the bell curve.

I will use the diagram from Wikipedia, with proper licensing evidence attached.

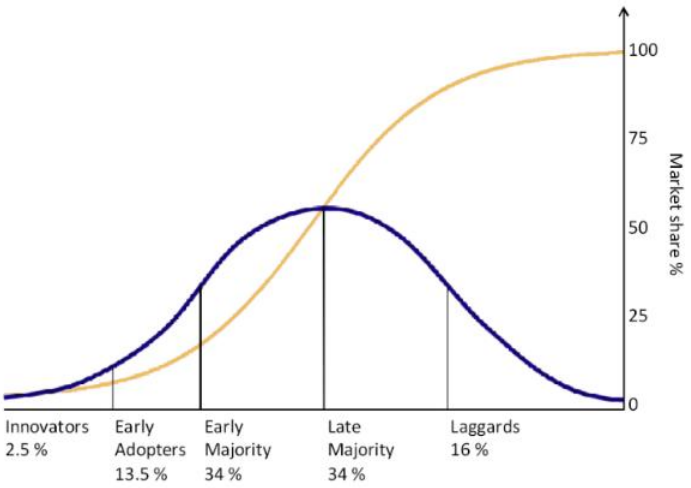


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However, the vertical axis on the right side of that diagram is labeled “market share”, and that is incorrect.

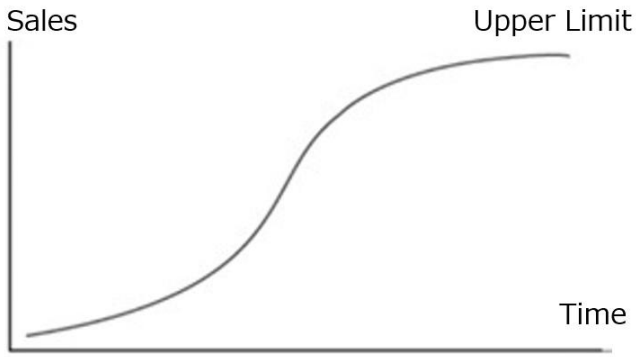
Rogers’ diffusion process describes the adoption of a single product, not market share. Naturally, there must be competing products of the same kinds (in-kind competitors) and also competing products of different kinds (not in-kind competitors).

This means that each competing product will eventually saturate according to its own Rogers diffusion curve. Therefore, the total market size—combining both in-kind and not-in-kind products—becomes the sum of multiple bell curves, each with different sizes.

The size of each individual bell curve represents the market share of that product. Up to this point, the logic should be clear.

Then, let me cite the description between pages 125 and 127 of my paper: "*The Mental Model of Innovation*".

3) Characteristics of the S-Curve



The growth curve has the following properties:

●If there is no replacement demand within a given period, the growth curve approaches the peak of the cumulative potential market size formed by products using similar or different technologies.

●Under monopoly conditions, the area under the curve—represented by a normal distribution whose upper limit is the asymptote of the S-curve—corresponds to the market size determined by the technology efficacy reflected in the product, expressed as a function of price.

●If all competing companies satisfy customers' purchasing-motivation factors equally (brand, delivery, dealer networks, advertising, etc.), then customer choice depends solely on the technology efficacy reflected in the product.

●Technology efficacy consists of four elements: performance, reliability, cost (not price), and usability.

Therefore, the upper limit is determined by the relationship between market needs for performance, reliability, and usability, and the price.

Here, usability refers only to size and mechanical complexity; design elements are excluded.

In short, the upper limit of the S-curve is logically determined by the technology efficacy reflected in the product as a function of price.
The slope is determined by market characteristics, maturity, entry barriers, and competitive marketing activities.

Among these properties, the first and second are easy to understand.
The third, however, has not been pointed out before.
It is unclear why this interpretation never appeared.
But the third and fourth properties make upper-limit inference possible.

4) Inferring the Upper Limit

- The three elements of technology efficacy can be evaluated relatively from the market's perspective.
  If similar products or services already exist, the estimation becomes easier.
  Cases with no competitors are extremely rare even with new technologies or new services.
  There is almost always a comparable market, product, or service.
- To infer the upper limit, focus only on the relationship between market needs at the time of commercialization and technology efficacy.
  Separate all marketing-strategy factors except price.
  Even if strategic elements slip into your thinking, try to separate them

as much as possible while moving to the next steps.

Step 1

Project your awareness forward to the point where the upper limit will eventually appear.

Compare the three technology-efficacy elements with both not-in-kind and in-kind competing products.

Remove desire, expectation, and bias as much as possible, and assign weights to the three elements as the market would see them.

At this stage, price does not need to be considered.

Step 2

During this comparison process, you will begin to sense the relationship between the price and sales volume of existing products.

Assume a price for the innovation product and repeat the technology-efficacy comparison.

5) Technology Efficacy Index

I propose a method for evaluating technology efficacy using two types of impact:

(1) the impact that enlarges the area of the normal-distribution curve **(market impact**), and

(2) the impact that does not enlarge the curve but still secures a certain market share (**market-share impact**).

Any measurement indicators may be used.

That's all. It's simple.

For source code, some people say the same four elements apply, while others recommend software-specific factors such as scalability, maintainability, technical debt, difficulty of imitation, or degree of coupling with data. I do not know which are appropriate.
You may, however, choose any set of technology-efficacy items that fit the logic of upper-limit inference presented here and are easy to use.

6) Four Core Skills needed for SEs in the Physical AI Era

While wandering around—trying to understand whether the "upper-limit search" for software's market value can be approached the same way as in manufacturing—I stumbled upon an extreme S-curve, shown here.

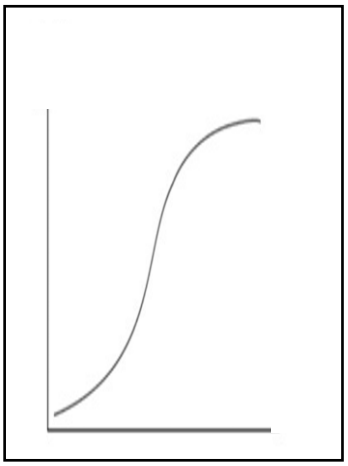


Figure 1: Global Trend in Software Spending, in USD Billion, Actual and Projected, 2019-2027
https://www.wipo.int/en/web/global-innovation-index/w/blogs/2025/global-software-spending?utm_source=openai

I asked Gemini: "Do you know why the curve flattened?"

(begin quote Gemini's answer)
From WIPO's data and market analyses by S&P Global, several concrete reasons can be identified:

1. The post-pandemic correction and the establishment of the 'new normal'

During the COVID-19 years (2020–2021), companies worldwide rushed to adopt remote work and shift to e-commerce, driving software investment up by 33% year-over-year. The flat trend after 2023 does not mean the market declined. Rather, the one-time surge settled into a high, elevated baseline—around USD 675 billion in 2024—where the market has stabilized.

2. A shift in investment toward AI physical infrastructure

Since the generative-AI boom of 2023, a large portion of IT budgets has moved away from software itself and toward the physical infrastructure required to run it: data centers, AI semiconductors (GPUs), servers, and networking equipment. S&P Global forecasts that massive capital expenditure in AI infrastructure is driving overall IT spending, while software's share is naturally leveling off.

(end quote)

I didn't copy the third reason—but the fourth one **clicked me** perfectly.

(begin quote Gemini's answer)

4. Efficiency gains in software development (no-code tools and AI-generated code)

As AI systems spread, the very nature of software development is changing. Code-generation tools and no-code AI builders are reducing the cost and time required to build systems that once demanded enormous development budgets. Companies can now create internal systems more quickly and at lower cost. (end quote)

When I asked, “If AI writes code, does that mean we’ll need fewer humans who write source code?”

(begin quote Gemini’s answer)
To answer the question about whether we still need humans who write source code:
the necessity of people who “simply write code as instructed”—coders—will decline sharply.
But the need for humans who design systems and solve problems—engineers—will only grow.

Four Core Skills Required for Engineers in the AI Era

Engineers who thrive in the AI era must go beyond memorizing programming languages.
They need to strengthen the following abilities:

- Requirements definition and communication

The ability to listen carefully, identify what the customer is truly struggling with, and express those needs clearly.

- AI-coordination skills (prompt engineering)

The ability to give precise instructions that enable AI to generate accurate and efficient code.

- System design and architecture selection

The ability to judge which technology stack should be used, considering scalability, maintainability, and long-term system health.

- Domain knowledge

Deep understanding of the rules and business models of the industry where the system will be used—healthcare, finance, logistics, and more.

Summary

In the coming era, the value of "being able to program" will decline.
The value of using technology to solve the problem in front of you will rise to its maximum.
Engineers who treat AI as a powerful subordinate—and position themselves as the director and architect—will be the ones overwhelmingly demanded by the market.
(end quote)

"Four Core Skills"—and again, **it clicked**. **Four times**.
What clicked?

This term!

In Chapter 34, "R&D: The Best is Business-Driven," from the 1992 book of Professor Drucker: ***Managing for the Future****,* there is a passage—and this term: **technology manager**.

(begin quote)
A Free-standing Business (p 218)
Bell Labs was founded to produce all the technology the telephone would require – and did so far for many years. Conversely, its parent company, AT&T, was expected to be able to use whatever new technology the lab would come up with. It did so for many years. But since World War II,

many of the technological breakthroughs to come out of Bell Labs – e.g., the transistor – have found their main application outside of their telephony. And more and more of the new telephone technology is coming from outside the telecommunication industry and its labs.
Siemens's one truly successful new product – the body scanner – was created not in its lab, but by a British producer of phonograph records.

The technological streams no longer run parallel. They increasingly cross each other, with frequent spillovers from one to the other.
Increasingly, therefore, the research lab may become a free-standing business, doing research work on contract for a multitude of industrial clients.

Each client would then need a '**technology manager**' rather than a 'research director' – someone who can develop business objectives based on the potential of technology and technology strategies based on business and market objectives, and who then defines and buys the technical work needed to produce business results. But no one today – and surely neither the engineering or business schools – knows how to teach technology management nor, indeed, even where to start.
(end quote)

And here is **the requirement for a technology manager**:
"A person who can define business objectives based on the potential of technology, formulate technology strategies based on those business and market objectives, identify the technical outcomes needed to produce

commercial results, and secure the people who can deliver those outcomes."

That's where it clicked.

The "Four Core Skills"required of future software engineers are essentially the very skills of a technology manager.

That realization was the spark.

So, I asked Gemini.

I'm not sure whether I should trust the answer, but—well—"**Yikes—didn't see that coming**."

7) Prof. Drucker's Refreshing Return after 34 Years

(begin quote Gemini's answer)

Professor Drucker's role of the technology manager can be broken into four steps.

These steps match perfectly, **one-to-one,** with today's "Four Core Skills" required for system engineers in AI robotics Era"

1. Define business objectives based on the potential of technology → **Domain knowledge**

To turn AI's potential into business objectives, you must understand the rules of the field—healthcare, finance, logistics. Industry problems and AI's possibilities meet here.

2. Formulate technology strategy based on business and market objectives → **System design / architecture selection**

Modern development is built by combining AI, no-code tools, and cloud services. Designing the optimal architecture for the business goal is the strategy itself.

3. Identify the technical outcomes needed to produce commercial results → **Requirements definition / communication**

Customers often cannot express what they truly need. Through dialogue, the engineer uncovers the real problem and translates it into clear requirements that AI can execute.

4. Secure the people who can deliver those outcomes → **AI-coordination skills (prompt engineering)**

In Drucker's era, this meant managing talented researchers. Today, the strongest "team" is AI. Giving precise instructions and getting the intended technical output fast is modern execution.

Conclusion:

AI, cloud, and no-code now intersect exactly as Drucker predicted.

The act of writing code has become widely accessible, and the engineer's role has shifted—from a craftsperson to a technology manager who translates technology into business results and directs AI as a subordinate.

It took 34 years, but the world has finally caught up to Drucker's future.

(end quote)

The world finally caught up— did it really?

I took Professor Drucker's remark—"no one knows where to begin"—as a personal assignment.

I believed there had to be a way to develop such people.
The answer came in a 2012 paper titled *The Mental Model of Innovation*, originally based on a 2000 guideline for venture capital.

Here is the summary of that paper:
The study analyzed how leaders think and act during the early stages of innovation driven by technological seeds, and identified two mental models.
The first is the mental model for exploring the upper limit of the growth curve of products that could emerge from a technology seed.
The second is the mental model for building a database of market and competitive information, linked to the business and marketing strategy required for commercialization.
This database acts as a sharp filter for future changes in markets and competitors, enabling quick judgment and decision-making that accelerates the entire innovation process.
(end quote)

●The mental model for exploring upper limits can be trained as we discussed here.
●But the mental model for building a market-and-competition database is far more difficult. There is a training method, but personal sense and intuition play a major role.

8) The Figure-8 Process of the Strategy Triangle

You already know the Strategy Triangle. Its three points—Corporation, Customer, Competitor—form the well-known 3C framework for strategy building.

I'll use it here to explain why "building a market-and-competition database is not that simple."

Innovation begins with no product.

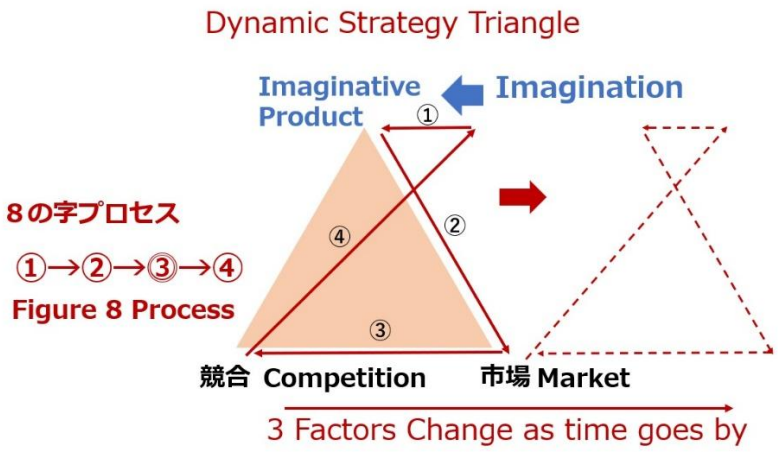
All you have is a technology seed and an idea.

The product exists only in your imagination.

And because it's imaginary, you tend to think only of the good parts.

You imagine a product, wonder whether it can be developed, imagine a market for it, and then try to see whether it could beat current and future competitors. When you run once around the Strategy Triangle—drawing a figure-8—you quickly realize how naive your first imagination was.

Sorry for the crude diagram.



So, you imagine a different product and repeat the same loop:

"What kind of product would actually work?"

Your image changes. Then, you repeat market fit and competitor comparison. As you repeat the loop, development issues gradually become visible, and you eventually reach a product concept that feels “about right.

”What matters is that, at this point, the strengths and weaknesses of your imagined product—against the market and against competitors—become concrete. And this is where the real problem lies.

People naturally avert their eyes from weaknesses because they want the project to survive.

But reducing weaknesses is strategy.

The more you run the figure-8, the more you must think—not just about the product, but about whether marketing strategy can turn weaknesses into strengths. At this stage, you’ve already begun tackling the difficult parts of development. You’re also thinking about market size—price and volume—and competitor strategy.

Even though the product is not yet fixed, you think that far ahead.

That’s why the upper-limit search must be done early.

Without it, you cannot build the mental database needed for strategy.

The absolute requirement for an innovation leader is to build this database early—based on the figure-8 process.

You estimate market size, meaning tentative revenue, and consider investment efficiency.

You don’t just think—you actually calculate profit and loss.

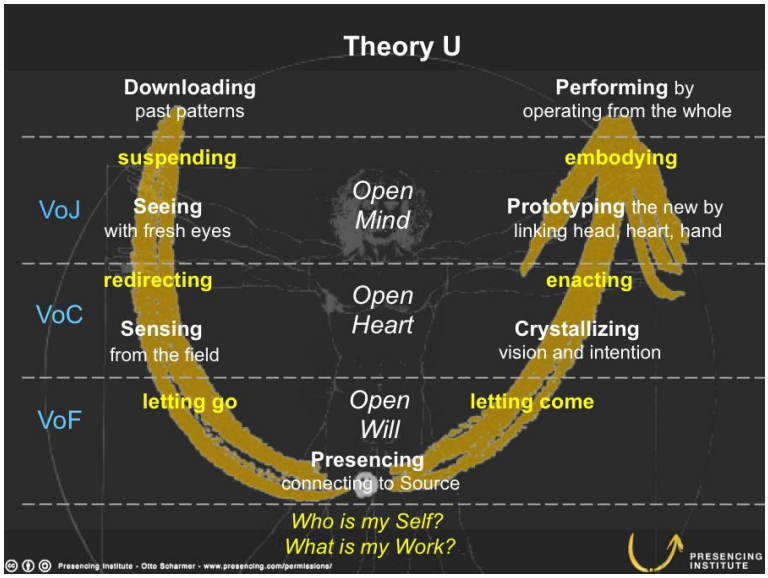
Numbers may change as development progresses, or they may not.

For manufacturing cost, you ask the production team for their sense of it—and you push until you get a solid answer.

Markets change. Competitors change. Development changes.
Unless the basis of your database is clear, you cannot respond to those changes. I'm not exaggerating.
At this stage, you do everything—absolutely everything—so that nothing is missed.
If something is missed, development and strategy must go backward.
If you go backward after a year, commercialization will be delayed two or three years. By then, the market and competitors will have changed.
You must never go backward.
Depending on the product, if commercialization is five years away, you still do all of this in the first year or so.
Everyone who has studied management knows the Strategy Triangle.
But almost no one repeats the figure-8 process consciously.
So their database is unstructured, and its basis is vague.
Such a database is useless.
The more you repeat the loop, the more you realize how demanding the work is. It takes time.
You gather research materials, analyze them, and consider other people's views.
Whether innovation succeeds depends almost entirely on this work.
Another problem is that it's hard to divide this work among multiple people. The leader must gather data that fits the leader's own scenario—the strategy story inside their head. Sensitivity to information depends on whether you already have that scenario.

I emphasize here that a very small team, including its leader, must do all of this work by themselves. Even within a small team, it is extremely difficult to share the leader's strategy scenario—the story inside the leader's head.

This is precisely where Theory U resonates.



https://commons.wikimedia.org/wiki/File:Theory_U.png

There is an important passage by Dr. Otto Scharmer on pages 142–143 of the Japanese Essential Edition of Theory U. He cites Francisco Varela's comment on the experiment conducted by Hubel and Wiesel with kittens.

(begin quote)

Newborn kittens were paired, one strapped onto the back of the other. Only the bottom kitten could move its legs. Although both kittens experienced the same physical space, only the bottom kitten developed normal vision. The top kitten, which merely "experienced movement" without engaging its own body, remained nearly blind.

Varela's point was clear: perception cannot be passive.
One must engage one's whole body. Much of today's institutionalized learning—and many innovation practices—resembles the top kitten. We outsource the very work we ought to be doing ourselves. That may be acceptable for simple problems. But a true innovator would never delegate the act of igniting the learning journey. Since such connections bring the seeds of the future into the world, one must venture there personally.
(end quote)

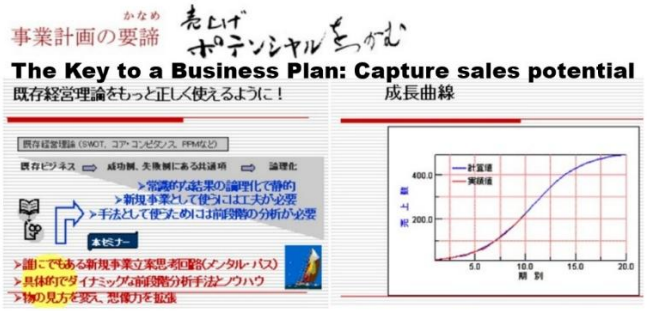
You cannot afford consultants at such an early stage anyway. But the kitten story is far more serious. We must never allow children growing up in the era of generative AI and physical AI to become "kittens at the top."

9) Humble suggestions to readers of Theory U

Dr. Otto Scharmer, a senior lecturer at the MIT Sloan School of Management, published the first edition in 2006: it is a massive work.
An English-language "Essentials" edition based on the original was published in 2016, followed by a Japanese edition in 2019.
On page 42 of the Japanese edition, Dr. Scharmer recounts, twenty years ago, when he began asking what was required to "learn from the future as it emerges," many people stared at him blankly. Many even told him that the question itself was misguided.

The concept that innovation begins by "exploring the upper limits of market value for technological seeds" does not involve the grand notion

of learning from an emerging future as Theory U envisions. However, it shares the same perspective: that the starting point of one's thinking lies in the future—specifically, at the stage of commercialization.



In a webinar I released in 2005 titled "How to Capture Sales Potential," I described this perspective as aligning with Professor Drucker's view of the Technology Manager.

I had assumed this perspective was common knowledge among those involved in innovation, but hearing Dr. Scharmer's account made me realize that, in reality, it was not.

While Theory U and the Figure 8 process of "Strategy Triangle" share this future-oriented starting point, they diverge critically when it comes to the path forward. Theory U is a masterpiece of personal and organizational transformation—focusing on "being" (how leaders alter their inner awareness). It invites us to descend into the "U" to let go of old mental models and listen to a collective source.

However, for a business innovation to survive, a shift in "awareness" is not enough. This is where Theory U leaves a massive void. It provides a profound philosophy for finding inspiration but lacks a concrete framework for constructing strategy. In the cold reality of the market, simply waiting for an "emerging future" to reveal itself is a recipe for being overtaken by faster, more ruthless competitors.

Of course, I respect Theory U very much because it handles much larger social and cultural objectives of human kind rather than just business innovation as read in page 30 of PDF version published in 2006.

(begin quote)
I personally believe that the current global situation yearns for a shift of the third kind, which in many ways is already in the making. We need to let go of the old body of institutionalized collective behavior in order to meet and connect with the presence of our highest future possibility.
The purpose of this book, and of the research and actions that have led to it, is to delineate a social technology of transformational change that will allow leaders in all segments of our society, including in our individual lives, to meet their existing challenges. In order to rise to the occasion, leaders often have to learn how to operate from the highest possible future, rather than being stuck in the patterns of our past experiences. Incidentally, when I use the word "leader," I refer to all people who engage in creating change or shaping their future, regardless of their formal positions in institutional structures. This book is written for leaders and change activists in corporations, governments, not-for-profit organizations, and communities. I have been often struck by how creators and master practitioners operate from a deeper process, one I call the "U Process." This process pulls us into an emerging possibility and allows us to operate from that altered state rather than simply reflecting on and reacting to past experiences. But in order to do that, we have to become aware of a profound blind spot in leadership and in everyday life.
(end quote)

I know that it may be a little too much to point out two problems I feel about Theory U, given Dr. Scharmer's noble mission.
However, I would like to do so for the sake of developing Drucker's "Technology Manager".

There is the U shape in page 43 for five movements from the top of the lefthand side of U and the second movement is:

• **Co-sensing:** go to the places of most potential; observe, observe, observe; listen with your mind and heart wide open.

If you exercise the dynamic Figure 8 process, you may not need to go down to the third movement, the bottom of U for **Co-presencing**.
Because You have already started from the future, the upper limit of the possible growth curve and created most probable strategy to achieve the future.

Furthermore, in order to strengthen your probable strategy, you have another powerful technology for deductive inference of competitors' strategies.

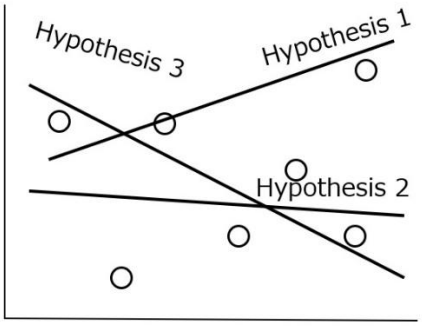


Fig. 1: Multiple hypothetical scenarios from random information

I. You piece together fragmentary information gathered from mainstream media and other open literatures and construct various scenarios based on how you would act in that situation. These become your hypotheses regarding the other party's strategic scenarios. These hypotheses need not be perfect; if the scenario hypothesis is a regression line, the fragmentary information represents the individual data points—the dots.

II. These hypotheses serve as a filter for selecting or discarding subsequently obtained information indicated by the black dots in Figure 2.

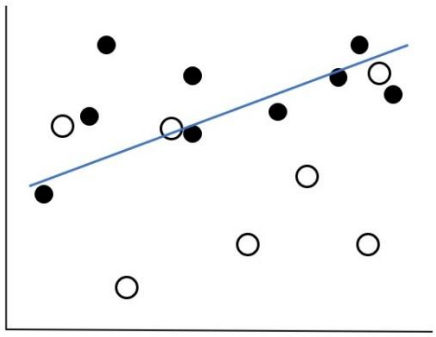


Fig. 2: Most likely strategic scenario

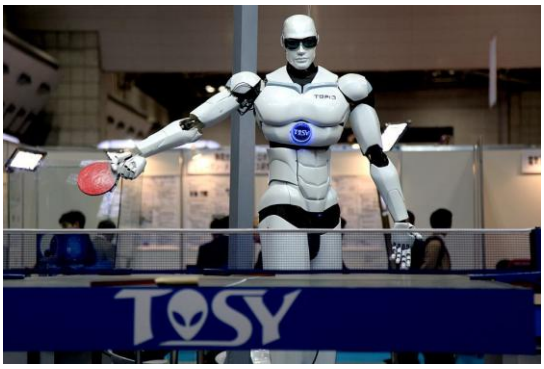
As a result, a regression line—or "story"—backed by numerous pieces of fragmentary information emerges. This line may share the same slope as the hypothesis as identical to it, or the slope may shift indicating a different story.

This becomes the counterpart's most likely strategic scenario. I call this method the “regression line method“.

In fact, this idea came from the cognitive science paper “Naive Causality: A Mental Model Theory of Causal Meaning and Reasoning” by Eugenia Goldvarg and P. N. Johnson-Laird, published in 2001.

10) Who on Earth is this guy, Hideo Watanabe?

Seventeen years ago—back in 2009—this image of a ping-pong-playing robot appeared:



TOPIO ("TOSY Ping Pong Playing Robot") is a bipedal humanoid robot designed to play table tennis against a human being.
https://commons.wikimedia.org/wiki/File:TOPIO_3.jpg

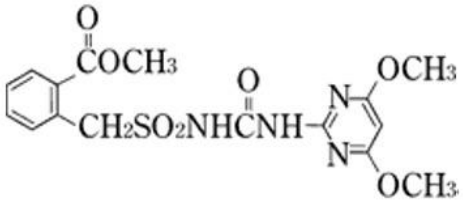
Japan's Ministry of Economy, Trade and Industry (**METI**) predicts that the global market for **AI robotics** will reach 60 trillion yen by 2040.
Building on Japan's long history of leadership in industrial robotics, METI has even declared an ambitious goal: a 30% share of the global market.

As I mentioned at the beginning, Japan certainly has "three-sigma people" like Mr. Furomuda.
But the true capability of Japan's IT industry is still unclear to me.

"Physical AI" may become the opportunity Japan needs to finally break out of its 35 lost years. That is why I decided to reveal everything I have learned about innovation—so that people in both Japanese manufacturing and IT industry can understand what innovation really requires.

I want them to see how essential the upper-limit search and database construction truly are.

Hideo Watanabe led the Japan-U.S. development team at DuPont from 1981 to 1987. The team developed the rice-field herbicide DPX-84 marketed as Londax.



It was a large project, and it was unusual for a foreigner to lead development. For DuPont, developing a herbicide for rice fields was a first. But before joining DuPont, I had spent eight years in a Japanese corporation working in marketing for rice-field herbicides. I knew the market, and I understood the technology to some extent.
It was not my intention but it happened to be in that position to lead the project because I believe I explored the upper limit of DPX-84's market value very early in the process. We commercialized DPX-84 in 7.5 years after the compound's invention—compared with the industry average of 11 years—and reached the upper limit in the third year after launch, following a perfect logarithmic curve instead of an S-curve.

Innovation is a human drama.

I recorded every drama happened during the development process of DPX-84 in my paper.

11) Path Forward

If the upper-limit search method has been useful to you, please share your own method for searching the upper limit of source code.
You can leave it as a comment for this booklet.

This is not a clumsy belief but a famous phrase of late Dr. Arnoldo Hax: “Strategy is not war with our competitors; it is love with our customers”.

Depending on your reactions, I may publish the Japanese version, including the human dramas of innovation—not only success stories but also failures caused by clinging to old mental models that Theory U profoundly implicates.

The Japanese version will also be DRM-free, so if you are interested in those dramas, I would like you to copy Japanese text and read them in your own language.

Professor Drucker passed away on November 11, 2005, at the age of 95.
Perhaps it’s a coincidence—but I was born on November 11.
Well, maybe it doesn’t matter.

Thank you very much for reading and best wishes for all of you.